TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	77.9 ± 3.62
Correlation Stability	80.0 ± 0.45
Distribution Consistency	82.9 ± 3.67
Entropy Preservation	85.9 ± 1.30
Mean Z Score Drift	0.3 ± 0.06
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	88.1 ± 2.55
Novelty Score	0.9 ± 0.01

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Dataset Audit: hepatitis.csv

1. Dataset Portrait

Attribute	Value
Rows	155
Columns	20
Numeric features	20
Categorical features	0
Missing cells (%)	5.4%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	41.200	32.718	-8.482	20.6%	Warn
Sex	1.103	1.097	-0.006	0.6%	OK
Steroid	1.506	1.524	+0.018	1.2%	OK
Antivirals	1.845	1.831	-0.015	0.8%	OK
Fatigue	1.351	1.403	+0.053	3.9%	OK
Malaise	1.604	1.597	-0.007	0.4%	OK
Anorexia	1.792	1.742	-0.050	2.8%	OK
Liver Big	1.828	1.790	-0.037	2.0%	OK
Liver Firm	1.583	1.548	-0.035	2.2%	OK
Spleen Palpable	1.800	1.879	+0.079	4.4%	OK
Spiders	1.660	1.645	-0.015	0.9%	OK
Ascites	1.867	1.871	+0.004	0.2%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	77.9 ± 3.62
Correlation Stability (*2)	80.0 ± 0.45
Distribution Consistency (*3)	82.9 ± 3.67
Entropy Preservation (*4)	85.9 ± 1.30
Mean Z Score Drift (*5)	0.3 ± 0.06
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00
Feature Accuracy (*10)	88.1 ± 2.55
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.867741935483871	0.8096774193548386	0.8483870967741935	-0.019354838709677413	
F1	0.5933333333333333	0.01818181818181818	0.464997224997225	-0.12833610833610834	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.8032258064516128	0.8741935483870968	0.03548387096774195	
F1	0.5132478632478632	0.08636363636363636	0.5795238095238096	0.06627594627594627	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	0.7967741935483871	0.8064516129032258	0.02580645161290325	
F1	N/A	0.0	0.02	0.02	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	0.5967741935483871	0.8032258064516128	0.0129032258064516	
F1	0.561449871078354	0.17575958092087124	0.5604444444444445	-0.00100542663390961 19	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	0.7064516129032259	0.8483870967741935	-0.00322580645161291 15	
F1	0.5385593354014405	0.013333333333333332	0.529577922077922	-0.00898141332351859 4	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.7967741935483871	0.7032258064516613	0.7709677419354839	-0.025806451612903226	
F1	0.4814141414141414	0.12522810522810524	0.3773382173382173	-0.10407592407592406	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.861290322580645	0.767741935483871	0.832258064516129	-0.029032258064516137	
F1	0.6062575106053367	0.11969696969696968	0.4651437451437451	-0.14111376546159154	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8064516129032258	0.7193548387096775	0.8064516129032258	2.2204460492503132e-17	
F1	0.3668398268398268	0.19537684537684535	0.4076098901098901	0.04077006327006327	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.7258064516129032	0.8548387096774193	0.01612903225806452	
F1	0.5718392134181608	0.03891402714932127	0.5144200244200244	-0.05741918899813636	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8064516129032258	0.8064516129032258	0.8064516129032258	0	
F1	N/A	0.0	N/A	0	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
2	50.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
8	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
9	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
10	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
11	39.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0
12	32.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
13	41.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
14	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
15	47.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
16	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
17	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
18	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
19	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
20	38.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	50.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0	2.0
2	18.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
3	5.0	1.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0
4	20.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	1.0	1.0
5	35.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
6	26.0	1.0	2.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
7	61.0	1.0	1.0	2.0	1.0	1.0	2.0	nan	nan	2.0
8	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
9	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
10	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	39.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	36.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0
3	22.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
4	37.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	1.0
5	46.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	72.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
7	60.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
9	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	20.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	8.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
3	63.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	28.0	1.0	2.0	2.0	1.0	1.0	2.0	1.0	2.0	2.0
5	29.0	2.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
6	26.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	19.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0
2	16.0	2.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
3	28.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
4	16.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
5	26.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
6	52.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
7	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
9	25.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
10	38.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	35.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0	1.0
2	27.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
3	43.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
4	15.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	47.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	45.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	1.0
7	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	46.0	1.0	2.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0
2	69.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	17.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
4	46.0	1.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0
5	60.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0
6	35.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
7	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
8	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
9	51.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
10	40.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	45.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
2	76.0	1.0	2.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
3	55.0	1.0	2.0	2.0	1.0	2.0	1.0	1.0	2.0	1.0
4	61.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0	2.0
5	50.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0
6	58.0	1.0	2.0	2.0	1.0	1.0	1.0	1.0	1.0	2.0
7	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
9	30.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	18.0	1.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0
2	34.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
3	35.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0
4	34.0	1.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0
5	28.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
6	20.0	1.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0
7	65.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
8	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
9	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
10	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	21.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0
2	28.0	1.0	1.0	2.0	2.0	2.0	2.0	1.0	1.0	2.0
3	24.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
4	44.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
5	19.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0
6	40.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	1.0
7	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	44.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
9	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	50.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
2	52.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	32.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
4	43.0	1.0	1.0	1.0	1.0	2.0	2.0	1.0	1.0	2.0
5	25.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
6	25.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0
7	56.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	28.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	49.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0